

# Few-shot Multimodal Anomaly Detection via Dynamic Intra-modal Sparsity Attention and Quality-aware Cross-modal Fusion in Microservice System

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## Abstract

The dynamic nature of microservice architectures necessitates robust few-shot anomaly detection (AD) systems to prevent cascading failures. However, current multimodal approaches remain inadequate for cold-start scenarios due to their dependence on extensive training data. The fundamental challenge in few-shot AD lies in two critical aspects: (1) effective feature extraction from intra-modalities, and (2) the alignment of heterogeneous multimodal representations, particularly the *distribution discrepancy* between discrete token embeddings from logs and continuously evolving representations from traces and metrics. To overcome these challenges, we present **FuseGuard**, an LLM-enhanced framework that pioneers quality-aware multimodal fusion for few-shot AD. First, our intra-modal feature extraction employs a Dynamic Sparse Correlation Attention mechanism for metrics, Spatio-temporal Encoding for traces, and LLM-enhanced semantic-frequency encoding for logs, enabling adaptive representation of time-evolving patterns. Second, we introduce hierarchical cross-modal alignment via token projection, layer-wise distillation, and consistency learning to preserve modality-specific features while enabling effective knowledge transfer. Finally, a quality-aware fusion mechanism dynamically weights modalities based on uncertainty estimation. Evaluated on three open-source platforms (MSDS, GAIA, TrainTicket) and a production system, FuseGuard's few-shot capability outperforms SOTA by 19.69%-77.07% F1-score across four datasets.

## CCS Concepts

• **Computing methodologies** → **Anomaly detection**; • **Software and its engineering** → *Software performance*.

## Keywords

Microservices Anomaly Detection; Multimodal Data Fusion; Cross-Modal Alignment

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## 1 Introduction

The evolution of microservices systems has provided a viable solution to address the growing scale and complexity of software systems. However, it has also amplified operational uncertainties [6]. Anomalies can escalate rapidly, leading to data loss, service outages, and financial impacts [1]. To address risks, major cloud platforms like AWS and Microsoft Azure now prioritize anomaly detection (AD) as a key safeguard to minimize the effects of anomalies [36].

To ensure effective AD, engineers predominantly rely on analyzing three data modalities: metrics (time-series data), logs (textual structured data), and traces (graph-structured data). Recent studies preprocess heterogeneous multimodal data to obtain feature matrix [17, 27, 28, 37] or unified fusion events [31, 34], and use them as inputs to deep learning or other machine learning methods. However, they fail to effectively address the critical challenge of the **cold-start scenario**, which requires reliable performance without extensive training samples.

The cold-start scenario in microservice AD necessitates few-shot learning capabilities due to the intrinsic dynamism of microservice ecosystems. Microservice systems experience continuous evolution through evolves, upgrades and routine maintenance[23], resulting in persistent pattern shifts[22], manifested as temporal distribution shifts across multimodal data streams. Current few-shot learning approaches [8, 32, 38] primarily concentrate on single-modal data analysis, which introduces substantial limitations. They may result in ineffective diagnosis due to the inability to capture or entirely miss abnormal information caused by certain failures.

Applying few-shot learning lies in effectively integrating multimodal data while maintaining rapid adaptability to distribution shifts. This problem can be decomposed into two core challenges:

- **Challenge 1: Sufficient exploitation on multimodal data.** Effective AD necessitates feature extractors capable of unifying heterogeneous data into a cohesive high-dimensional representation space to enable cross-modal fusion. **For metric modality**, modeling multivariate dependencies (e.g., CPU, memory, network) remains challenging. Independent modeling approaches [17, 34, 37] enhance noise robustness but lose cross-variable dependencies.

Full dependence approaches [11, 21, 27, 28, 31] capture interactions but risk spurious correlations in Table 1. Furthermore, existing approaches overlook frequency-domain analysis, which is critical for detecting anomalies manifested in periodic patterns and cross-variable spectral relationships. **For log modality**, current multimodal fusion approaches primarily incorporate log features at a superficial statistical level [11, 27, 31, 34, 37] (e.g., log frequency, temporal patterns), failing to leverage deeper semantic information. While these statistical methods can detect basic anomalies, they cannot fundamentally interpret contextual clues in logs (e.g., error stack traces, severity indicators).

- **Challenge 2: Robust multimodal alignment and fusion.** The core difficulty stems from the inherent disparity between modalities: while metrics and traces exhibit precise temporal structures but limited semantics, logs offer rich semantic information with loose temporal organization. Current methods suffer limitations: Feature concatenation [16] obscures critical temporal patterns (e.g., periodicity, anomaly pulses) and disrupts coherence, while attention mechanisms [17, 21, 28] (Table 1) indiscriminately blend modalities, letting textual noise corrupt temporal representations. Further, fine-grained temporal patterns in metrics/traces misalign with log tokens. Additionally, significant quality discrepancies exist among modalities due to transmission errors or noise. Consequently, multimodal analysis should incorporate considerations for both modal alignment and quality assessment.

To address such issues, we introduce powerful Large Language Models (LLMs) [29] in multimodal AD, leveraging robust embeddings learned from abundant language data to facilitate few-shot learning. We propose **FuseGuard**, a novel framework that combines dynamic feature learning with quality-aware multimodal alignment and fusion. **Challenge 1: Multimodal Feature Extraction** is addressed through FuseGuard’s novel architecture comprising three specialized components: a dynamic sparse correlation attention mechanism for metrics that performs frequency-domain feature extraction via soft-clustered sparsification; a spatio-temporal encoder for traces that simultaneously captures topological dependencies and temporal dynamics, and an LLM-enhanced semantic-frequency encoder for logs that preserves both contextual semantics and statistical features. The dynamic sparse attention employs adaptive soft clustering to identify and emphasize predictive variable relationships while suppressing noise, maintaining sparsity constraints through learnable frequency-domain filters. This tri-encoder design enables precise intra-modal feature extraction while preserving modality-specific characteristics essential for downstream multimodal fusion. **Challenge 2: Cross-Modal Alignment and Fusion** is addressed through a three-tiered framework: (1) Cross-modal token alignment establishes a unified representation space through nonlinear projections of heterogeneous features, (2) Layer-wise feature distillation enables progressive knowledge transfer between modalities via hierarchical attention mechanisms, and (3) Semantic consistency learning aligns output distributions to resolve inter-modal discrepancies. The framework culminates in FuseGuard’s quality-aware fusion module, which leverages energy-based uncertainty estimation to dynamically adjust modality weights. The end-to-end architecture simultaneously resolves: pattern shift through

**Table 1: Compared to FuseGuard, existing methods did not comprehensively address the multimodal AD challenge (Freq for Frequency and Sem for semantic).**

| Approach         | Few-shot learning | Single modal Feature Extraction |     | Cross-modal Fusion  |                        |
|------------------|-------------------|---------------------------------|-----|---------------------|------------------------|
|                  |                   | Freq                            | Sem | Extraction strategy | Fusion      Align      |
| Hades            | ×                 | ×                               | ✓   | Independence        | Cross-attn    ×        |
| MISP             | ×                 | ×                               | ✓   | Full Dependence     | Cross-attn    ×        |
| Nezha            | ×                 | ×                               | ×   | Independence        | ×              ×       |
| AnoFusion        | ×                 | ×                               | ×   | Independence        | ×              ×       |
| MSTGAD           | ×                 | ×                               | ×   | Full Dependence     | ×              ×       |
| ART              | ×                 | ×                               | ×   | Full Dependence     | ×              ×       |
| Medicine         | ×                 | ×                               | ✓   | Full Dependence     | Channel-attn   ×       |
| MRCA             | ×                 | ×                               | ×   | Full Dependence     | ×              ×       |
| <b>FuseGuard</b> | ✓                 | ✓                               | ✓   | <b>Dynamic Corr</b> | <b>Quality-Aware</b> ✓ |

adaptive sparse attention, modality gaps via hierarchical alignment, and noise interference using quality-aware fusion.

We conducted extensive studies to evaluate FuseGuard on four open-source platforms: MSDS [24], GAIA [4], TrainTicket [39], and a real-world production microservice system. FuseGuard’s few-shot capability redefines multimodal learning, with 10% training data, it outperforms SOTA by 19.69%-77.07% F1-score across four datasets. In summary, this paper has made the primary contributions:

- We are the first to achieve few-shot learning capability through effective intra-modal feature extraction and multimodal alignment in microservice environments.
- We present the first quality-aware multimodal fusion addressing real-world modality quality discrepancies, enabling adaptive noise suppression and optimal information integration.
- Extensive experiments on four datasets demonstrate that FuseGuard outperforms state-of-the-arts.

## 2 Motivation

FuseGuard leverages **Dynamic Intra-modal Feature Extraction** and **Cross-Modality Alignment** to effectively establish multimodal data reciprocal relationships for few-shot learning. This section mainly explains why these two methods are used.

### 2.1 Why Dynamic Intra-modal Feature Extraction?

Existing intra-modal feature extraction strategies face fundamental limitations in modeling multivariate metric modality. As illustrated in Figure 1(a-c), current approaches typically adopt one of three strategies: 1) Independent treatment of each variable [17, 34, 37], 2) Fully dependent modeling of all variables [11, 21, 27, 28, 31], or 3) Stable clustering of variables [20]. While independent strategies provide robustness through shared model parameters, they completely ignore potentially valuable inter-variable interactions, limiting their generalization capability. Dependent strategies, though capable of generating joint representations, often suffer from noise propagation through irrelevant channel correlations. Stable cluster strategies attempt to balance these trade-offs through hard clustering, but their rigid partitioning of variables restricts modeling flexibility and fails to capture dynamic relationships. More critically, our analysis in Figure 1(e) indicates that correlation coefficients

between metrics vary significantly over time, highlighting the limitations of fixed interaction patterns. This inspires our dynamic approach, which enables adaptive soft clustering of variables to emphasize valuable inter-variable relationships while minimizing irrelevant noise.

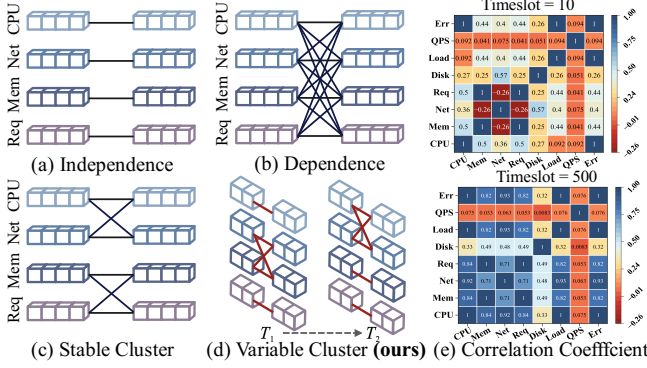


Figure 1: (a-d) Different feature extraction strategies. Squares represent different variables (CPU, Mem, Net, etc.) in the metric modality. (e) The heatmap shows significant changes in correlation coefficients between variables over time.

## 2.2 Why Cross-Modality Alignment?

Existing approaches for multimodal AD analysis face critical limitations that necessitate effective modality alignment. As shown in Figure 2(a), single-modality models [8, 33] either produce semantically weak time series embeddings or irrelevant text representations. While direct concatenation [16] (Figure 2(b)) attempts to combine modalities, it disrupts time series structure through uncontrolled mixing. More sophisticated multimodal approaches [17, 21, 28] (Figure 2(c)) still suffer from persistent entanglement that introduces textual noise into time series representations. Our proposed cross-modality alignment Figure 2(d) addresses these issues by aligning the structural integrity of time series data with the semantic robustness of language models. Figures 2 (e) and (f) visually confirm that our method provides superior embedding cohesion compared to MRCA [31] based on cross-modality attention, underscoring the importance of modality alignment in learning joint representations across temporal and textual modalities.

## 3 Proposed Approach

Figure 3 presents the overall architecture of FuseGuard, which consists of *intra-modal feature extraction*, *cross-modal alignment*, and *quality-aware multimodal fusion*. The framework first performs modality-specific feature extraction: capturing dynamic correlations for metrics, spatiotemporal patterns for traces, and both semantic and statistical features for logs. To bridge the modality gap between traces, metrics, and logs, we employ cross-modal fine-tuning of pre-trained LLMs. Finally, the system dynamically adjusts modality weights based on real-time quality assessment for optimal fusion. Next, we elaborate on these stages.

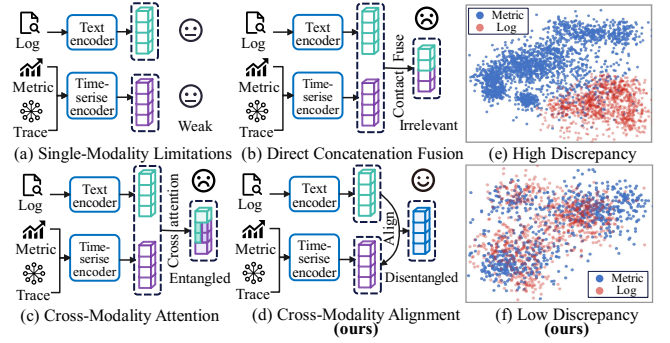


Figure 2: (a) Single-modality limitations: text encoders generate time series-irrelevant representations. (b) Concatenation issues: direct fusion mixes modalities and disrupts time-series structure. (c) Multimodal Entanglement: joint embeddings remain entangled with textual noise. (d) Cross-Modality alignment: improve semantic consistency between textual (log) and temporal modalities (metric and trace). (e)-(f): T-SNE comparison, our method (f) shows tighter clustering than MRCA [31] (e), demonstrating better alignment.

### 3.1 Intra-modal Feature Extraction

**3.1.1 Dynamic Sparse Correlation Attention for Metrics.** To address issues from inaccurate modeling of inter-metric correlations among microservices, we propose a metric learning framework that dynamically clusters metrics in the frequency domain. This approach generates a learnable channel mask matrix, preserving essential dependencies while reducing noise and redundancy.

**Learnable Frequency Domain Distance.** To enhance metrics through their interdependencies, we propose modeling their correlations in the frequency domain. This approach captures richer patterns by breaking down signals into sinusoidal components using Fourier bases. Specifically, given a discrete time series of the  $i$ -th metric,  $H_i^M = \{h_{i,0}^M, \dots, h_{i,T}^M\} \in \mathbb{R}^{F_M \times T}$ , where  $T$  denotes the sequence length and  $F_M$  the total number of metrics, we first project it into frequency domain using the real-valued Fast Fourier Transform (rFFT) [2], thus, we have  $\hat{H}_i^M = \text{norm}(\text{rFFT}(H_i^M)) \in \mathbb{R}^{F_M \times T/2}$ . Next, we derive a learnable Mahalanobis distance matrix [7] in frequency domain to evaluate inter-metric relationships:

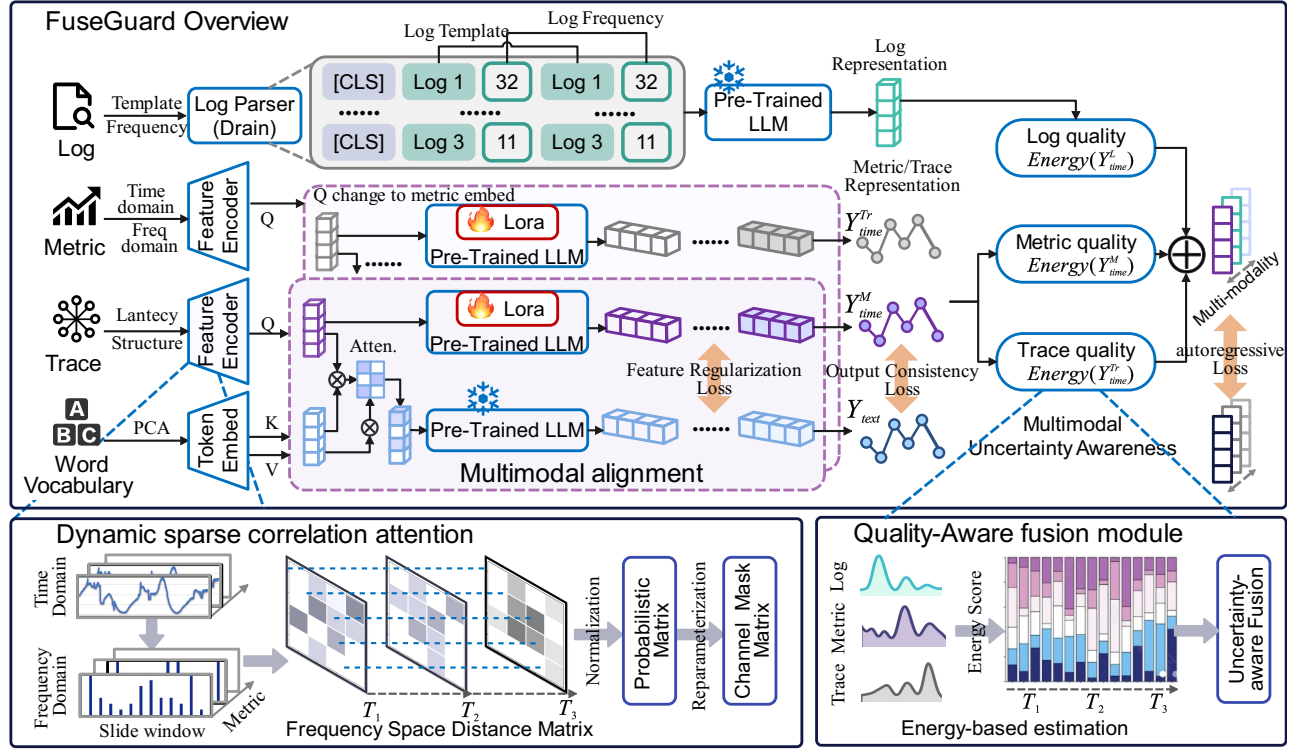
$$d(H_i^M, H_j^M) = (\hat{H}_i^M - \hat{H}_j^M)^T \cdot Q \cdot (\hat{H}_i^M - \hat{H}_j^M) \quad (1)$$

we introduce a learnable positive semi-definite matrix  $Q \in \mathbb{R}^{\frac{T}{2} \times \frac{T}{2}}$ , constructed as  $Q = A^T \cdot A$  (where  $A$  is a trainable parameter matrix).

**Probabilistic Correlation Normalization.** Leveraging the learned distance matrix  $D_{ij} = d(H_i^M, H_j^M)$ , we first compute the relationship matrix  $Re_{ij} \in [0, 1]$  of metrics and normalize it. We then apply an element-wise transformation to  $Re_{ij}$  to obtain a probability matrix  $P_{ij}$  using a discount factor  $\gamma \in (0, 1)$ , which preserves discriminative information while suppressing outliers:

$$Re_{ij} = \begin{cases} \frac{1}{D_{ij}} & \text{if } i \neq j \\ 0 & \text{if } i = j \end{cases}, P_{ij} = \begin{cases} \frac{Re_{ij} \cdot \gamma}{\max_j(Re_{ij})} & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases} \quad (2)$$

$P_{ij}$  represents the probability that metric  $j$  is useful for metric  $i$ . **Reparameterized Sampling.** To selectively preserve beneficial



**Figure 3: Framework Overview.** *FuseGuard* employs a three-stage multimodal diagnostic pipeline: (1) **Intra-modal feature extraction**, where the metric branch utilizes *Dynamic Sparse Correlation Attention* to capture evolving variable dependencies; (2) **Cross-modal Alignment**, featuring fine-tuned LLMs that achieve token-level synchronization to bridge the modality gap; and (3) **Quality-aware Fusion**, which dynamically weights modalities through energy-based reliability estimation.

inter-metric relationships while filtering out detrimental ones, we employ the probability matrix  $P_{ij}$  to guide a Bernoulli-sampled binary mask  $M \in \mathbb{R}^{F_M \times F_M}$ , where each element  $M_{ij} \approx \text{Bernoulli}(P_{ij})$ . For differentiable training, we implement the sampling process using the Gumbel-Softmax reparameterization trick [13], which maintains gradient flow by the discrete sampling operation.

**Feature extraction.** Finally, we extract temporal features for the metric modality using the mask matrix  $M_{ij}$  through a masked attention mechanism, where  $Q = H_i^M \cdot W^Q$ ,  $K = H_i^M \cdot W^K$ ,  $V = H_i^M \cdot W^V$ .  $W^Q, W^K, W^V \in \mathbb{R}^{d \times d}$  are projection matrices in the attention block. The architecture incorporates layer normalization, feed-forward networks, and skip connections, following the standard Transformer paradigm. This process can be formally expressed as:

$$\begin{aligned} Msk &= \frac{Q \cdot K^T}{\sqrt{d}} \odot M + (1 - M) \odot (-\infty) \\ Em^M &= \text{Softmax}(Msk) \cdot V, \end{aligned} \quad (3)$$

where  $Msk \in \mathbb{R}^{F_M \times F_M}$  is the attention score matrix and  $Em^M$  is the metric feature.

**3.1.2 Spatio-temporal Encoding for Traces. Temporal Dynamics Modeling.** In a microservice system, traces capture user request executions as spans, which include attributes like trace ID, request type, span ID, and duration. Analyzing these spans sequentially helps us extract structural information and dependencies. We can

also convert duration data into a time series by calculating the average latency per minute. The  $i$ -th span is represented as  $H_i^{Tr} = \{h_{i,0}^{Tr}, \dots, h_{i,T}^{Tr}\} \in \mathbb{R}^{F_{Tr} \times T}$ , where  $F_{Tr}$  is the number of spans. To capture latency patterns and temporal fluctuations, we apply 1D temporal convolutions with a sliding kernel across the time axis. The temporal convolution for span  $i$  is computed as:

$$z_{i,t}^{(l+1)} = \sum_{k=0}^{K-1} \Theta_k^{(l)} h_{i,t-k}^{(l)}, \quad (4)$$

where  $\Theta_k^{(l)}$  are the learnable filters of kernel size  $K$ . Stacked temporal convolutions with dilated kernels help model long-range dependencies in latency time series.

**Spatial Dependency Modeling.** In a Spatio-temporal graph, nodes represent spans, and edges represent scheduling relationships between different spans. We apply graph convolution to aggregate neighboring span features. For the  $i$ -th span at time step  $t$ , the graph convolution operation is defined as:

$$h_{i,t}^{(l+1)} = \sigma \left( \sum_{j \in \mathcal{N}(i)} \frac{1}{c_{ij}} W^{(l)} h_{j,t}^{(l)} + b^{(l)} \right), \quad (5)$$

where  $\mathcal{N}(i)$  denotes the neighbors of span  $i$ ,  $W^{(l)}$  and  $b^{(l)}$  are learnable parameters at layer  $l$ . We finally obtain the trace modality feature  $Em^{Tr}$  by stacking multiple Spatio-Temporal Blocks.

**3.1.3 LLM-enhanced Semantic-Frequency Encoding for Logs.** We use Drain3 [28] to convert unstructured system logs into structured log messages represented as log templates. The log data is divided into fixed-size time windows, and for each window, we create a log sequence that contains all unique log templates and their corresponding occurrence frequencies. To capture these patterns, we include frequency information with each log template, providing additional features for identifying potential system failures. The resulting log sequences are input into LLM to generate embedded representations that combine the semantic information of the log templates and their frequency patterns. This creates a log-modality feature,  $Em^L$ , which encompasses both the structural and behavioral aspects of the system logs.

### 3.2 Cross-modal Alignment

As illustrated in Figure 3, after extracting features from multimodal data, we first perform *Cross-modal Token Alignment* to achieve token-level spatial alignment between the temporal modalities (metrics, traces) and textual embeddings (word vocabulary). Next, we employ *Layer-Wise Feature Distillation*, where  $L$  stacked pre-trained LLM layers are utilized to derive intermediate representations for both temporal and textual modalities. Finally, *Semantic Consistency Learning* ensures that the output distributions of the temporal and textual modalities remain aligned.

**3.2.1 Cross-modal Token Alignment.** To mitigate the distributional discrepancy between temporal (metrics, traces) and textual tokens, we propose to project the temporal modality into the LLM’s textual embedding space. The input distribution of the pre-trained LLM is defined by its word embedding matrix  $\mathcal{V}oc \in \mathbb{R}^{|\mathcal{A}| \times F}$ , where  $|\mathcal{A}|$  denotes vocabulary size and  $F$  the hidden dimension. Directly computing cross-attention between time-series tokens and  $\mathcal{V}oc$  is computationally prohibitive (since  $|\mathcal{A}|$  can reach tens of thousands).

To address this, we introduce a principal word embedding extraction strategy, leveraging the observation that semantically similar words form synonym clusters. Specifically, we reduce the dimensionality of  $\mathcal{V}oc$  via Principal Component Analysis (PCA), retaining only the top  $a$  principal components ( $a \ll |\mathcal{A}|$ ), yielding a compact embedding matrix  $\hat{\mathcal{V}oc} = \text{PCA}(\mathcal{V}oc) \in \mathbb{R}^{a \times F}$ .

We then employ Multi-head Cross-Attention, using  $\hat{\mathcal{V}oc}$  as keys/values and the temporal features of metrics  $Em^M$  and traces  $Em^{Tr}$  as queries, respectively, to align the principal word embeddings with time-series tokens. This produces aligned textual tokens  $Em^{\mathcal{V}oc}$ :

$$Em^{\mathcal{V}oc} = \text{Softmax}\left(\frac{QK^T}{\sqrt{E}}\right)V, \quad (6)$$

$$Q = EmW_q, K = \hat{\mathcal{V}oc}W_k, V = \hat{\mathcal{V}oc}W_v,$$

where  $W_q, W_k$  and  $W_v \in \mathbb{R}^{M \times M}$  are the projection matrices for the query  $Q$ , key  $K$ , and value  $V$ , respectively. When the temporal modality is metric,  $Em = Em^M$  and  $E = F_M$ ; When the temporal modality is trace,  $Em = Em^{Tr}$  and  $E = F_{Tr}$ .

**3.2.2 Layer-Wise Feature Distillation.** The pre-trained weights in LLMs are optimized for their original textual modality. To better adapt these weights to temporal modalities (metrics, traces), we align the intermediate features between temporal and textual modalities through a feature regularization loss. This alignment process

enables more effective gradient propagation across all transformer layers, leading to improved weight updates. Formally, let  $F_l^{text}$  and  $F_l^{time}$  denote the intermediate features from the  $l$ -th transformer block of the textual and temporal modalities, respectively. The feature regularization loss is defined as:

$$\mathcal{L}_{feature} = \sum_{i=1}^L \gamma^{(L-i)} \text{sim}(\phi_i^{text}(F_{text}^l), \phi_i^{time}(F_{time}^l)), \quad (7)$$

where  $\gamma$  is a hyperparameter that controls the loss scale from different layers, and  $\text{sim}()$  is a chosen similarity function.

**3.2.3 Semantic Consistency Learning.** We improve semantic consistency between textual (log) and temporal modalities (metric and trace) using an output consistency loss. This function aligns output distributions, resolving discrepancies and ensuring unified representations for both data types, which enhances model prediction accuracy and reliability. Formally, given the outputs  $Y_{text}$  (textual modality) and  $Y_{time}$  (temporal modality), the output consistency loss is defined as  $\mathcal{L}_{output} = \text{sim}(Y_{text}, Y_{time})$ .

### 3.3 Quality-Aware Multimodal Fusion

To minimize the impact of low-quality modalities, we propose an uncertainty-aware fusion mechanism that dynamically adjusts integration weights based on modality-specific uncertainty.

**3.3.1 Modality-wise Uncertainty Estimation.** We establish an uncertainty-aware fusion framework by equating modality quality with predictive reliability, quantified through an energy-based uncertainty measure. For multimodal data, the density functions  $\log p(Y_{time}^{(m)})$  of  $m$ -th modalities can be estimated by the corresponding energy function:

$$\log p(Y_{time}^{(m)}) = -\frac{\text{Energy}(Y; G^m)}{T^m} - \log Z^m \quad (8)$$

where  $\text{Energy}()$  is the energy function.  $-\text{Energy}(Y; G^m)$  is linearly aligned with density  $\log p(Y_{time}^{(m)})$ . The energy score for the  $m$ -th modality of input  $Y_{time}^{(m)}$  can be calculated as

$$\text{Energy}(Y_{time}^{(m)}) = -\sum_{t=1}^T \mathcal{T}_t \cdot \log p(y_t | Y_{\leq t}^{(m)}) \quad (9)$$

where  $\mathcal{T}_t$  is a temperature parameter. Lower energy scores indicate high-confidence predictions, while higher scores reflect lower confidence and unreliable outputs.

**3.3.2 Uncertainty-aware Weight Fusion.** The energy scores are transformed into fusion weights enabling automatic suppression of high-uncertainty modalities during cross-modal integration. The fusion weight  $w^m(Y_{time})$  for modality  $m$  is computed as a linear function of its energy score:

$$w^m(Y_{time}) = \alpha^m \cdot \text{Energy}(Y_{time}^{(m)}) + \beta^m \quad (10)$$

with  $\alpha^m < 0$  to ensure that higher energy (i.e., higher uncertainty) reduces the modality’s influence. To satisfy the theoretical condition  $r(w^m, \ell^m) \leq 0$  (negative correlation between weights and losses), historical prediction errors can be used for regularization, such as via exponential moving averages or sliding-window loss tracking. Thus, multimodal fusion can be expressed as  $Y_{\text{fuse}} = w^M Y_{\text{time}}^M + w^{Tr} Y_{\text{time}}^{Tr} + w^L Y_{\text{time}}^L$ , where  $w^M + w^{Tr} + w^L = 1$ .

### 3.4 Anomaly Evaluation

To prevent catastrophic forgetting and improve training efficiency, we utilize a parameter-efficient training approach to fine-tune the pre-trained LLMs. Specifically, for the metric and trace branches, we implement Low-Rank Adaptation (LoRA) and modify the weights associated with the positional encoding. Given the ground-truth future sequence  $X_{t+1:t+H}$  over the next  $H$  steps, the autoregressive model conditioned on the fused embedding  $Y_{\text{fuse}}$  yields predictions  $\hat{X}_{t+1:t+H}$ . The loss is

$$\mathcal{L}_{AR} = \frac{1}{H} \sum_{\tau=1}^H \|\hat{X}_{t+\tau} - X_{t+\tau}\|_2^2.$$

The total loss during training is the weighted summation of the autoregressive loss  $\mathcal{L}_{AR}$ , the feature regularization loss  $\mathcal{L}_{\text{feature}}$ , and the output consistency loss  $\mathcal{L}_{\text{output}}$ :

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{AR} + \lambda_1 \mathcal{L}_{\text{feature}} + \lambda_2 \mathcal{L}_{\text{output}} \quad (11)$$

where  $\lambda_1$  and  $\lambda_2$  are hyperparameters.

We utilize the Continuous Ranked Probability Score (CRPS) [14] as the abnormal score for each instance during the testing process. CRPS quantifies the degree of compatibility between a cumulative distribution function (CDF)  $F$  and an observation  $x$  for each modal data serialization and is calculated as  $(F, x) = \int_{\mathbb{R}} (F(z) - \mathbb{I}\{x \leq z\})^2 dz$ . The abnormal score of  $i$ -instance can be calculated by summing the  $(F, x)$  of three modalities [5], resulting in  $\hat{F}_{\text{sum}}(t)$  for each time point. The findings are subsequently calculated as the average over the prediction horizon.  $AbScore = \mathbb{E}_t \left[ \text{CRPS} \left( \hat{F}_{\text{sum}}(t), \sum_{i=1}^D x_{i,t} \right) \right]$ .

Anomalies are observations that deviate from expected patterns in historical data. They occur when the expected conditional distribution probability of a prediction length  $T_p$  significantly differs from the actual value. FuseGuard calculates the anomaly severity using the  $AbScore$ . To trigger an anomaly event, it's crucial to select a threshold. In this study, we use a non-parametric, dynamic unsupervised threshold selection method as introduced in [12]. NDT first creates the set  $e$  given a set of abnormal scores  $s$  of historical observations, where  $z$  is an ordered set of positive values. Then, the threshold  $Th$  is calculated by

$$Th = \text{argmax}(e) = \frac{(\Delta\mu(s)/\mu(s)) + (\Delta\sigma(s)/\sigma(s))}{|s_a| + |E_{\text{sea}}^2|} \quad (12)$$

where  $\Delta\mu(s) = \mu(s) - \mu(s < Th)$  and  $\Delta\sigma(s) = \sigma(s) - \sigma(s < Th)$ , NDT also considers preventing generating a large number of anomalous values  $s_a = \{s > Th\}$  and anomalous sequences  $|E_{\text{sea}}^2|$ . NDT aims to identify a threshold that results in the most significant decrease in the average and variability of  $s$  by eliminating any values over the threshold.

## 4 Evaluation

### 4.1 Experiment Setting

**4.1.1 Dataset.** We conducted experimental research on both open-source datasets and datasets generated from real-world systems.

**Open-source Microservice System:** We utilize three open-source microservice benchmarks, namely MSDS [24], Generic AIOps Atlas (GAIA) [4], and TrainTicket [39] (TT) which has been widely used in the existing work [3, 11, 16, 18, 19, 35], The MSDS dataset is

**Table 2: Basic information about four datasets.**

| Dataset | Modules | Failure Types | Failures | the Size of Modality |             |            |
|---------|---------|---------------|----------|----------------------|-------------|------------|
|         |         |               |          | Metric               | Log         | Trace      |
| MSDS    | 5       | 3             | 5347     | 1,195,637            | 139,799     | 742,384    |
| GAIA    | 10      | 8             | 1099     | 2,623,580            | 87,974,577  | 58,684,253 |
| TT      | 41      | 3             | 143      | 1,570,266            | 108,637     | 148,296    |
| RMS     | 56      | 35            | 1117     | 6,122,354            | 214,546,769 | 67,533,723 |

produced from running a complex distributed system (Openstack). GAIA is collected from a system utilized by both mobile and PC users. TT is derived from an open-source microservice benchmark system for the train ticket service. More detailed dataset information is provided in Appendix A.

**Real-world Microservices-based System:** This system is a real-world production microservices system (RMS) with 56 microservices in a large Internet company with more than two million users (part of the system). The typical faults include the following: CPU stress, memory stress, high disk I/O latency, disk partition full, database failure, resource exhaustion, high packet loss rate, etc.

**4.1.2 Evaluation Metrics & Baselines.** To evaluate the accuracy of AD, following the common practice in AD methods [21, 26, 30, 37], we choose Precision, Recall, and F1 to evaluate the effectiveness of anomaly methods based on TP (True Positive), FP (False Positive), and FN (False Negative).

To comprehensively evaluate the AD performance of the model, we selected eight representative baselines for comparison: Hades [17], MISP [21], Nezha [34], AnoFusion [37], Medicine [28], MRCA [31], MSTGAD [11] and ART [27]. The detailed baseline models can be found in Appendix B.

**4.1.3 Implementation Details.** For the GAIA and TT datasets, we evaluated forecasting horizons of  $\{24, 36, 48, 80\}$ , while using extended horizons  $\{96, 192, 336, 720\}$  for the longer-sequence MSDS and RMS datasets, with empirically selected lookback windows of  $\{36, 100\}$  for GAIA/TT and  $\{96, 336, 512\}$  for others. Our architecture leverages the first 6 layers of a pre-trained GPT-2 model [25], fine-tuned via LoRA (rank = 8,  $\alpha = 32$ ) [10] with dropout ( $p = 0.1$ ) for robustness, optimized using Adam ( $\eta = 5 \times 10^{-4}$ ) [15] with a composite loss ( $\gamma = 0.8$ ,  $\lambda_1 = 1$ ,  $\lambda_2 = 0.01$ ). The initial batch size of 256 was dynamically halved (min = 8) to prevent OOM errors.

### 4.2 Performance Comparison

Our comprehensive evaluation demonstrates FuseGuard's superior performance against multiple baseline methods across four datasets (Table 3). The proposed method consistently outperformed all probabilistic baselines, achieving significant F1-score improvements of 3.73%, 3.45%, 3.39%, and 7.08% over the strongest competitor on each respective dataset. Experimental results reveal that bimodal methods generally underperform multimodal approaches due to their limited capacity for cross-modal analysis, while performance variations across datasets correlate strongly with their inherent complexity. Specifically, all methods achieved their best results on the TT dataset, which features simpler relationships and lower dimensionality, whereas the RMS dataset proved most challenging

**Table 3: The effectiveness of FuseGuard in comparison with baselines on four datasets. The 'Num' column shows the number of integrated modalities, with boldface indicating the best overall performance and underlining marking the strongest baseline.**

| Num   | Approach         | MSDS         |              |              | GAIA         |              |              | TT           |              |              | RMS          |              |              |
|-------|------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|       |                  | Precision    | Recall       | F1-score     | Precision    | Recall       | F1-score     | Precision    | Recall       | F1-score     | Precision    | Recall       | F1-score     |
| Two   | Hades            | 0.716        | 0.704        | 0.710        | 0.731        | 0.707        | 0.719        | 0.742        | 0.731        | 0.736        | 0.511        | 0.691        | 0.588        |
|       | MISP             | 0.761        | 0.731        | 0.746        | 0.768        | 0.742        | 0.755        | 0.791        | 0.753        | 0.772        | 0.573        | 0.531        | 0.551        |
| Three | Nezha            | 0.433        | 0.691        | 0.532        | 0.544        | 0.642        | 0.589        | 0.642        | 0.738        | 0.687        | 0.483        | 0.691        | 0.569        |
|       | AnoFusion        | 0.774        | 0.791        | 0.782        | 0.784        | 0.773        | 0.778        | 0.804        | 0.783        | 0.793        | 0.631        | 0.715        | 0.670        |
|       | Medicine         | 0.826        | 0.804        | 0.815        | 0.835        | 0.815        | 0.825        | 0.847        | 0.832        | 0.839        | 0.774        | 0.794        | 0.784        |
|       | MRCA             | 0.874        | 0.862        | 0.868        | 0.883        | 0.871        | 0.877        | <u>0.952</u> | <u>0.938</u> | <u>0.945</u> | 0.837        | 0.821        | 0.829        |
|       | MSTGAD           | <u>0.928</u> | <u>0.919</u> | <u>0.923</u> | 0.918        | 0.904        | 0.911        | 0.947        | 0.933        | 0.940        | 0.865        | <u>0.853</u> | <u>0.859</u> |
|       | ART              | 0.907        | 0.912        | 0.909        | <u>0.931</u> | <u>0.922</u> | <u>0.926</u> | 0.941        | 0.928        | 0.934        | <u>0.874</u> | 0.842        | 0.858        |
|       | <b>FuseGuard</b> | <b>0.964</b> | <b>0.952</b> | <b>0.958</b> | <b>0.968</b> | <b>0.949</b> | <b>0.958</b> | <b>0.982</b> | <b>0.972</b> | <b>0.977</b> | <b>0.925</b> | <b>0.912</b> | <b>0.918</b> |

**Table 4: Few-shot learning results on 20% and 10% training data of four datasets. Boldface denotes the best result and underline indicates the best result among baselines.**

| Approach         | Few-shot learning on 20% training data |               |              |               |              |               |              |               | Few-shot learning on 10% training data |               |              |               |              |               |              |               |
|------------------|----------------------------------------|---------------|--------------|---------------|--------------|---------------|--------------|---------------|----------------------------------------|---------------|--------------|---------------|--------------|---------------|--------------|---------------|
|                  | MSDS                                   |               | GAIA         |               | TT           |               | RMS          |               | MSDS                                   |               | GAIA         |               | TT           |               | RMS          |               |
|                  | F1                                     | Reduction     | F1           | Reduction     | F1           | Reduction     | F1           | Reduction     | F1                                     | Reduction     | F1           | Reduction     | F1           | Reduction     | F1           | Reduction     |
| Hades            | 0.631                                  | -11.12%       | 0.652        | -9.29%        | 0.684        | -7.12%        | 0.497        | -15.41%       | 0.602                                  | -15.21%       | 0.594        | -17.36%       | 0.653        | -11.33%       | 0.453        | -22.90%       |
| MISP             | 0.655                                  | -12.16%       | 0.674        | -10.70%       | 0.694        | -10.05%       | 0.501        | -9.11%        | 0.619                                  | -16.99%       | 0.621        | -17.72%       | 0.675        | -12.51%       | 0.472        | -14.37%       |
| Nezha            | 0.512                                  | -3.83%        | 0.572        | -2.88%        | 0.672        | -2.14%        | 0.529        | -6.96%        | 0.489                                  | -8.15%        | 0.533        | -9.50%        | 0.658        | -4.17%        | 0.508        | -10.65%       |
| AnoFusion        | 0.657                                  | -16.03%       | 0.642        | -17.53%       | 0.687        | -13.41%       | 0.569        | -15.12%       | 0.623                                  | -20.37%       | 0.611        | -21.51%       | 0.664        | -16.31%       | 0.541        | -19.30%       |
| Medicine         | 0.732                                  | -10.17%       | 0.715        | -13.32%       | 0.761        | -9.34%        | 0.678        | -13.51%       | 0.704                                  | -13.60%       | 0.695        | -15.75%       | 0.747        | -11.01%       | 0.622        | -20.65%       |
| MRCA             | 0.784                                  | -9.67%        | 0.773        | -11.85%       | <u>0.875</u> | -7.40%        | 0.709        | -14.47%       | 0.762                                  | -12.21%       | 0.733        | -16.42%       | 0.811        | -14.18%       | 0.669        | -19.29%       |
| MSTGAD           | <u>0.821</u>                           | -11.10%       | 0.803        | -11.85%       | 0.872        | -7.23%        | 0.719        | -16.29%       | <u>0.781</u>                           | -15.43%       | <u>0.785</u> | -13.83%       | <u>0.839</u> | -10.74%       | 0.657        | -23.51%       |
| ART              | 0.801                                  | -11.93%       | 0.787        | -15.05%       | 0.858        | -8.18%        | <u>0.724</u> | -15.59%       | 0.774                                  | -14.90%       | 0.749        | -19.16%       | 0.835        | -10.64%       | <u>0.702</u> | -18.15%       |
| <b>FuseGuard</b> | <b>0.947</b>                           | <b>-1.14%</b> | <b>0.943</b> | <b>-1.61%</b> | <b>0.971</b> | <b>-0.61%</b> | <b>0.903</b> | <b>-1.68%</b> | <b>0.933</b>                           | <b>-2.61%</b> | <b>0.927</b> | <b>-3.28%</b> | <b>0.959</b> | <b>-1.84%</b> | <b>0.891</b> | <b>-2.99%</b> |

due to its larger data volumes, complex microservice dependencies, and higher noise levels. Among the baselines, ART showed particularly strong performance on GAIA and RMS datasets by effectively capturing spatiotemporal features, while MSTGAD's semi-supervised approach delivered stable but suboptimal results, performing relatively better on the spatially simpler TT dataset. MRCA's VAE-based architecture demonstrated advantages in modeling temporal dynamics and uncertainty. FuseGuard's consistent outperformance across all datasets stems from its novel dual-phase architecture, which combines dynamic sparse correlation analysis for robust intra-modal feature extraction with quality-aware dynamic multimodal fusion and explicit cross-modal alignment, enabling comprehensive analysis of complex multimodal patterns in microservice systems.

### 4.3 Few-shot Learning

Few-shot learning capability has become a crucial requirement for microservice AD. To rigorously evaluate our method's generalization capacity, we design comprehensive few-shot learning experiments where only limited portions (10% and 20%) of the training data are available. This challenging setting particularly tests a model's ability to extract meaningful patterns from severely constrained information.

We evaluate on four benchmark datasets under these few-shot conditions. As demonstrated in Table 4, our method consistently outperforms all baseline approaches, achieving average reductions of 2.61%, 3.28%, 1.84%, and 2.99%, respectively, across different data regimes. This superior performance can be attributed to FuseGuard's novel multimodal framework that effectively combines LLM fine-tuning with cross-modal learning. The results validate that FuseGuard's cross-modal learning mechanism enables robust feature extraction and knowledge transfer even with scarce training data, leading to more accurate predictions compared to conventional approaches. This advantage makes our method particularly suitable for real-world scenarios, particularly cold start scenarios triggered by microservice modifications.

### 4.4 Ablation Study

**Loss Function Analysis.** We conduct a systematic evaluation of FuseGuard's loss components through three ablated variants:

- **w/o L\_feature+L\_output:** Retains only the supervised loss for direct autoregressive learning.
- **w/o L\_feature:** Removes the feature regularization loss that aligns intermediate representations between textual (log) and temporal (trace/metric) feature extractors.
- **w/o L\_output:** Eliminates the cross-modal output consistency.

**Table 5: Ablation on different loss and module functions.**

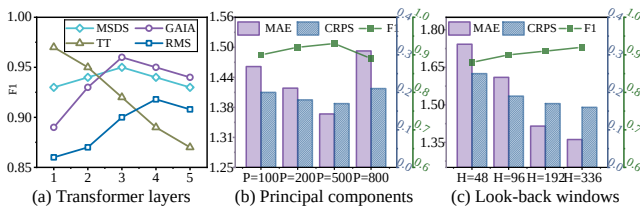
| Ablation | Variant                    | MSDS<br>F1 | GAIA<br>F1 | TT<br>F1 | RMS<br>F1 |
|----------|----------------------------|------------|------------|----------|-----------|
| Loss     | A.1 w/o L_feature+L_output | 0.853      | 0.825      | 0.849    | 0.772     |
|          | A.2 w/o L_feature          | 0.847      | 0.831      | 0.855    | 0.789     |
|          | A.3 w/o L_output           | 0.875      | 0.849      | 0.886    | 0.808     |
| Module   | B.1 w/o Dynamic sparse     | 0.903      | 0.906      | 0.931    | 0.852     |
|          | B.2 w/o Modal alignment    | 0.864      | 0.834      | 0.858    | 0.850     |
|          | B.3 w/o Quality-Aware      | 0.923      | 0.894      | 0.955    | 0.788     |

We analyze the specific effects of each proposed loss function as detailed in Table 5. The addition of feature regularization loss or output consistency loss led to incremental improvements, with the best performance observed when all three losses were combined, achieving the highest F1-score on four datasets.

**Architectural Module Analysis.** We further investigate three critical architectural modifications:

- **w/o Dynamic sparse:** Replaces dynamic sparse attention with full attention
- **w/o Modal alignment:** Removes cross-modal feature alignment.
- **w/o Quality-Aware:** Removes the quality assessment module.

Dynamic Sparse Attention (B.1) proves most crucial for the RMS dataset with substantial temporal distribution shifts. The masked attention mechanism reduces model complexity and effectively filters irrelevant inter-channel correlations present in full attention. Cross-Modal Alignment (B.2) demonstrates universal importance across all datasets. This component bridges the distribution gap between heterogeneous modalities (textual logs vs. temporal metrics), enabling effective information fusion. Quality-Aware Fusion (B.3) offers notable benefits in production environments, especially in the RMS dataset. This is mainly because production data can be affected by transmission errors and environmental interference, which may introduce noise. By using multimodal quality assessment, lower weights are assigned to low-quality modalities for fusion.

**Figure 4: Parameter sensitivity analysis.**

## 4.5 Hyperparameters Sensitivity

**Effect of transformer layers.** The ideal number of Transformer layers depends on the dataset complexity. For simpler datasets like TT, one layer is sufficient to avoid overfitting. In contrast, more complex datasets like RMS require multiple layers to capture intricate feature interactions, ensuring a balanced model capacity.

**Effect of principal components.** We enhance training efficiency by using Principal Component Analysis to reduce the dimensionality of word embeddings. While PCA decreases computational costs, it can lead to information loss. We assess different numbers of principal components,  $P$ . As shown in Figure 4(a), model performance remains stable with varying  $P$ . However, too few components can harm performance by losing important information, while too many can introduce redundancy and complicate learning. Therefore, we set  $P = 500$  to maintain satisfactory performance.

**Effect of look-back window.** The length of the look-back window is critical to forecasting performance: a window that is too short risks overlooking salient temporal patterns, whereas one that is too long may inject noise or obsolete information. Remarkably, our results reveal a sustained performance gain as the look-back window lengthens, underscoring the model's ability to effectively capture long-range temporal dependencies.

## 4.6 Interpretability of Failure Diagnosis

**Solution to Challenge 1.** Our dynamic sparse attention module captures inter-metric correlations among key metrics, such as Memory status, Memory limit, Memory usage, Memory failure, and Network drop. As illustrated in Figure 5(a), it learns frequency-domain relationships through masked attention weights, creating adaptive clusters while maintaining important cross-group connections. This approach effectively preserves critical dependencies and filters out noise through sparse attention.

**Solution to Challenge 2.** To bridge the gap between modalities, our cross-modal alignment module converts raw logs into semantically aligned embeddings using attention mechanisms. Figure 5(b.1) shows cross-attention maps highlighting key patterns. Technical terms like "Pattern" and "Status" align well with metrics, averaging an attention weight of 0.08, while general terms such as "World" show much less engagement (weight  $< 0.02$ ). Additionally, the diagonal patterns in the attention distributions indicate a precise match between metric patterns and their textual descriptions.

The effectiveness of our quality-aware fusion is shown in Figure 5(b.2 and b.3.1-b.3.4), highlighting the adaptive weighting in four scenarios: (1) Balanced cases with equal contributions from all modalities (log, metric, trace) at  $0.33 \pm 0.02$ ; (2) Metric-depressed Fusion, where the metric contributes minimally ( $0.21 \pm 0.04$ ); (3) Trace-depressed Fusion, with reduced influence from the trace ( $0.19 \pm 0.05$ ); and (4) Log-depressed Fusion, where the log contributes the least ( $0.22 \pm 0.03$ ). The t-SNE visualizations confirm that high-quality modalities cluster tightly, while noisy inputs are more dispersed as the red dashed circles. This adaptability is beneficial in production environments with varying modality quality.

## 5 Related Work

Anomaly detection in microservice systems is a complex yet critical task, leading to a wide array of proposed solutions. To enhance the applicability of anomaly detection across a broader range of failures, the integration of multimodal data has emerged as a crucial priority in both academic and industrial settings. Multimodal data provides a comprehensive representation of the system's state, significantly improving the reliability and interpretability of diagnostics [11, 17, 37]. Effective anomaly detection typically requires the inclusion

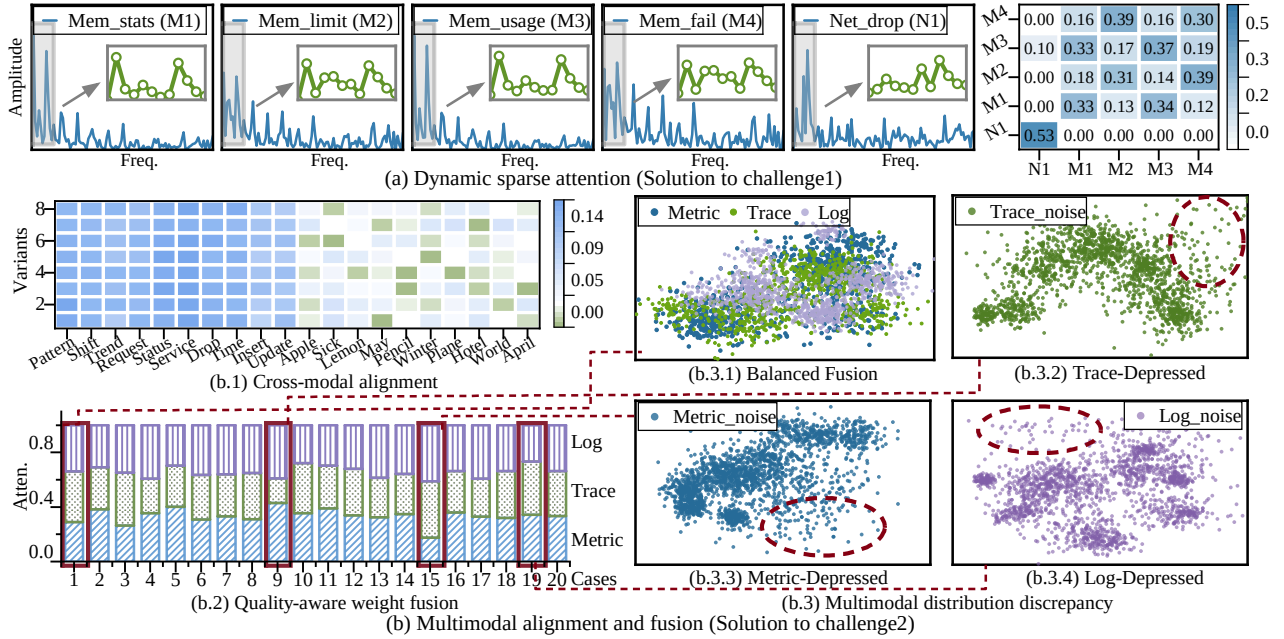


Figure 5: FuseGuard’s effectiveness in handling challenging anomaly evaluations.

of at least two out of three data types: logs, metrics, and traces. Multimodal fault diagnosis techniques can be categorized into three main approaches: result fusion, model fusion, and feature fusion.

**Result Fusion.** Result fusion techniques typically begin by performing anomaly detection or similar operations on individual data modalities. The final failure detection is determined by either correlating abnormal scores with predefined measures or using a voting mechanism to combine results from multimodal processing [9]. For instance, PDiagnose [9] is a pioneering study that leverages logs, metrics, and traces for fault diagnosis, demonstrating the effectiveness of result fusion in integrating diverse data sources.

**Model Fusion.** Model fusion methods often require more complex designs due to the need to handle multiple data modalities simultaneously. Graph-based models, such as AnoFusion [11, 37], excel at integrating multimodal data by storing feature information in graph nodes and employing autoregressive prediction to generate diagnostic results. However, achieving high accuracy with these models often incurs significant training costs, making them resource-intensive.

**Feature Fusion.** Recent research has focused on transforming multimodal data into a unified feature representation. This process begins with preprocessing operations to extract fundamental information from diverse data types, resulting in a feature matrix [17, 28] or a consolidated fusion event [34]. Feature fusion approaches aim to create a unified representation while maintaining interpretability. These unified features are then used as inputs for deep learning or other machine learning techniques, enabling the extraction of critical multimodal feature information and improving diagnostic accuracy.

## 6 Conclusion

The dynamic and evolving nature of microservice architectures necessitates robust anomaly detection mechanisms capable of handling multimodal data under cold-start scenarios, where traditional approaches often fall short due to insufficient training samples and ineffective cross-modal fusion. To bridge this gap, we propose FuseGuard, a novel few-shot learning framework that redefines multimodal anomaly detection through dynamic feature extraction, hierarchical alignment, and quality-aware fusion. By integrating LLM-enhanced semantic analysis with sparse frequency-domain attention and uncertainty-weighted multimodal fusion, FuseGuard achieves remarkable performance gains—outperforming state-of-the-art methods by up to 77.07% in F1-score with only 10% of training data. Extensive evaluations on real-world and open-source microservice platforms, including MSDS, GAIA, TrainTicket, and production systems, demonstrate FuseGuard’s superior ability to capture cross-modal dependencies, adapt to distribution shifts, and optimize fusion reliability.

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## A Dataset Information

We deploy three open-source microservice benchmarks, three open-source microservice benchmarks, namely MSDS [24], Generic AIOps Atlas (GAIA) [4], and TrainTicket [39] (TT).

- **MSDS.** The MSDS dataset is a multi-modal collection of distributed traces, logs, and system metrics from an Open-Stack testbed deployed using Kolla-Ansible in a Dockerized microservice environment. Workloads and realistic fault injections were generated using Rally, including os-faults to simulate production-like anomalies. The dataset covers 5 physical nodes with 7 metrics (e.g., CPU/RAM usage), 23 log features, and trace attributes (e.g., service name, span ID).
- **GAIA.** The GAIA is created in a simulation that includes 10 microservices, two database services (MySQL and Redis), and five host machines. This system serves mobile and PC users. Operators introduced five types of failures, which include system failures like system freezes and process crashes, and service failures such as login failures, missing files, and access denials.
- **TT.** TrainTicket is a microservice-based railway ticketing system that simulates real-world operations, allowing users to check, book, and pay for train tickets. The dataset captures interactions among 41 microservices, with 27 dedicated to core business logic (e.g., bookings, payments, user management). We construct a distributed testbed to deploy the systems running in Docker containers and develop two request simulators to simulate valid user requests.

## B Baseline Comparison

To comprehensively evaluate the AD performance of the model, we selected eight representative baselines for comparison: Hades [17], MISP [21], Nezha [34], AnoFusion [37], Medicine [28], MRCA [31], MSTGAD [11] and ART [27]. It is important to note that the original method is supervised in the Medicine [28] baseline. We retained the steps for multimodal processing and feature extraction in the original method and replaced the final fully connected layer with an unsupervised clustering approach. The anomaly detection models adopted by each method are as follows:

- **Hades** utilizes a hierarchical architecture to learn a global representation of the system status by combining log semantics and metric patterns. It captures significant features and meaningful interactions from diverse data through a cross-modal attention module.
- **MISP** transforms multimodal data into multi-dimensional sequences, extracting and encoding features both within and across modalities, ultimately computing the failure probability from the synthesized features.
- **Nezha** converts diverse multi-modal data into a unified event representation and identifies event patterns by constructing and analyzing event graphs.
- **AnoFusion** utilizes a Graph Transformer Network (GTN) to learn the correlations of heterogeneous multimodal data and integrates a Graph Attention Network (GAT) with a Gated Recurrent Unit (GRU) for dynamically changing multimodal data.
- **MSTGAD** constructs a transformer-based neural network incorporating both spatial and temporal attention mechanisms to model the inter-correlations among different modalities and the temporal dependencies between data points.
- **ART** sequentially models channel, temporal, and call dependencies using a Transformer Encoder, GRU, and GraphSAGE, respectively. Unified failure representations enhance the interpretability of features.
- **Medicine** encodes different modalities separately to retain their unique features and employs adaptive optimization to adjust the learning pace between modalities, thereby enhancing overall diagnostic performance.
- **MRCA** utilizes traces and logs to obtain the ranking list of abnormal services based on reconstruction probability. It further builds causal graphs from services with high anomaly probability to discover the order in which abnormal metrics of different services occur.

A comparative literature review reveals that FuseGuard presents several key innovations: (1) For **Challenge 1**, we are the first to jointly leverage the frequency characteristics of time-series modality (metrics and traces) and the semantic features of textual modality (logs), employing dynamic correlation for effective intra-modal feature extraction. (2) For **Challenge 2**, we propose the first quality-aware dynamic multimodal fusion framework with explicit cross-modal alignment, addressing inherent heterogeneity in multimodal integration.